Enterprise Infrastructure Plan

ABC Startup Solutions

ITEP 414 – System Administration and Maintenance

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Week 2 Portfolio Project | 26 October 2026

This professional planning document defines the first IT environment for a 20-person software-development startup. It is intended for management review before procurement and deployment.

Executive Summary

ABC Startup Solutions has no current computers, servers, network, internet infrastructure, or security policies. This plan provides a supportable small-business design with secure departmental segmentation, managed endpoints, protected storage, documented backup, and capacity for early growth.

Part 1 — Company Profile

Item	Profile
Company	ABC Startup Solutions
Nature of business	Custom web and mobile software development services.
Vision	To be a trusted Philippine software partner that delivers useful, secure digital products.
Mission	To build reliable client solutions through collaborative teams, secure engineering practices, and responsive support.
Office	2nd Floor, Laguna Tech Hub, Santa Cruz, Laguna, Philippines (fictional).
Structure	Managing Director → department leads → IT, HR, Finance, and Sales staff.

Employee distribution: IT 5; HR 4; Finance 5; Sales 6; total 20 employees. The design keeps room for 10–15 additional users.

Part 2 — Enterprise Hardware Inventory

Asset	Hardware	Qty.	Department	Purpose / justification
HW-001	Business desktop; Core i5/Ryzen 5, 16 GB RAM, 512 GB SSD	18	HR, Finance, Sales	Standardized office endpoints plus three spares.
HW-002	Business laptop; Core i7/Ryzen 7, 16 GB RAM, 1 TB SSD	5	IT	Administration, development, VMs, and mobility.
HW-003	ECC server with RAID storage and remote management	1	IT	Identity, files, printing, monitoring, and internal test VMs.
HW-004	24-inch IPS monitor	23	All	One per endpoint for productivity and ergonomic use.
HW-005	Network multifunction laser printer	2	Shared	Separate locations reduce queues; secure print can be controlled centrally.
HW-006	1500 VA UPS	2	Server/network	Safe shutdown and outage protection.
HW-007	4-bay NAS, 16 TB usable RAID	1	IT	Local snapshots and backup target.
HW-008	Encrypted external drive, 8 TB	2	IT	Offline/off-site rotated backup copies.

Part 3 — Enterprise Software Inventory

Software	Version / edition	License	Purpose
Windows 11 Pro	Current supported	Commercial	Business endpoint OS with BitLocker and domain/Entra support.
Ubuntu Server	24.04 LTS	Open source	Linux services, monitoring, and internal tools.
Microsoft 365 Business Standard	Current	Subscription	Email, collaboration, Office apps, cloud files.
VS Code / Git / GitHub Desktop	Current stable	Free / GPL	Editing and versioning scripts, code, and documentation.
VirtualBox	Current stable	GPLv3	Isolated test virtual machines.
Chrome	Current stable	Freeware	Managed access to SaaS and documentation.
Defender for Business	Current	Plan included	Endpoint protection and alerts.
AnyDesk / 7-Zip	Current stable	Business / LGPL	Audited support and archive handling.

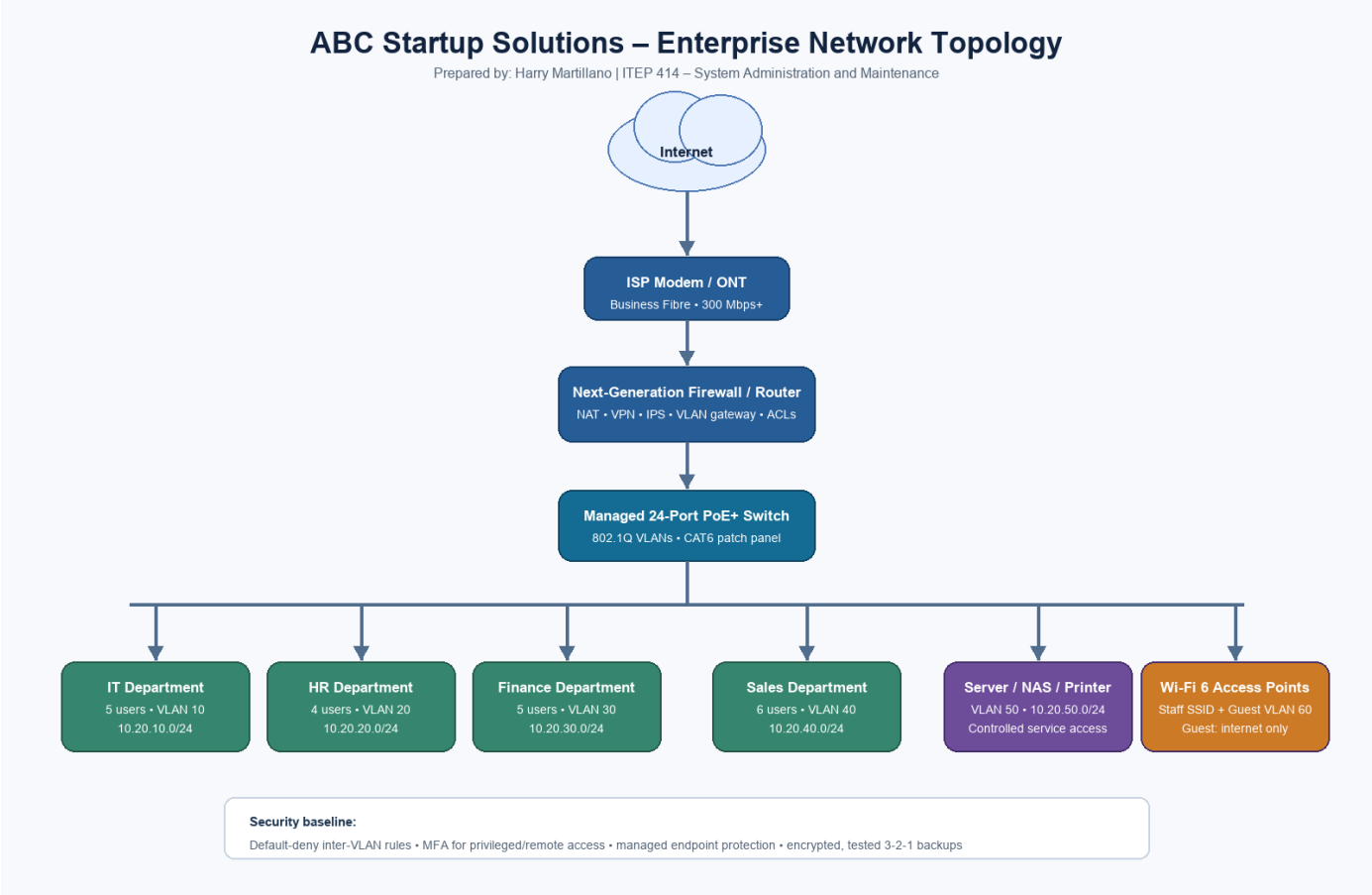
Part 4 — Enterprise Network Inventory

Asset	Equipment	Qty.	Specification / role
NET-001	Business fibre modem/ONT	1	Terminates a minimum 300 Mbps symmetrical circuit.
NET-002	Next-generation firewall/router	1	NAT, VPN, VLAN routing, IPS, filtering, logging, and policy enforcement.
NET-003	Managed 24-port Gigabit PoE+ switch	1	VLAN-capable access/core; PoE powers access points.
NET-004	Wi-Fi 6 access point	2	Staff coverage plus isolated guest SSID.
NET-005	24-port CAT6 patch panel	1	Maintainable structured-cabling termination.
NET-006	CAT6 runs	30	Desk, printer, AP, and rack cabling with spares.
NET-007	RJ45 connectors/keystones	40	Terminations and contingency stock.

Network Segmentation Plan

VLAN	Network	Users / devices	Access rule
10	10.20.10.0/24	IT	Named administrator access to infrastructure.
20	10.20.20.0/24	HR	HR services and approved internet.
30	10.20.30.0/24	Finance	Finance services; deny other user VLANs by default.
40	10.20.40.0/24	Sales	Sales services and approved internet.
50	10.20.50.0/24	Server/NAS/printers	Explicit firewall rules only.
60	10.20.60.0/24	Guest Wi-Fi	Internet only; no internal access.

Part 5 — Enterprise Network Diagram



The firewall is the policy boundary. The managed switch carries tagged VLANs to department endpoints, shared services, and access points. Inter-VLAN traffic is denied by default and only required services are allowed.

Part 6 — System Administration Roles

Helpdesk Technician

Resolves user incidents, prepares devices, documents tickets, and escalates useful diagnostics. Skills: customer communication, Windows troubleshooting, endpoint basics, and documentation. Tools: service desk, remote support, Microsoft 365 administration, endpoint console. Certifications: CompTIA A+, ITIL Foundation.

Network Administrator

Designs and maintains LAN, Wi-Fi, VPN, routing, switching, firewall rules, and monitoring. Skills: TCP/IP, DNS, DHCP, VLANs, wireless, and change management. Tools: firewall/switch dashboards, Wireshark, monitoring. Certifications: Network+, CCNA, vendor firewall credentials.

Linux System Administrator

Installs, hardens, patches, automates, monitors, and recovers Linux services. Skills: permissions, storage, systemd, SSH, logging, Bash, backups, and automation. Tools: Bash, SSH, Ansible, Git, journalctl. Certifications: Linux+, LPIC-1, RHCSA, LFCS.

Cloud Administrator

Manages identities, cloud accounts, resources, networking, storage, security, costs, backups, and availability. Skills: IAM, virtual networking, automation, monitoring, governance, and response. Tools: Azure, AWS, CLI, Terraform. Certifications: AWS Cloud Practitioner/SysOps Associate, Azure Administrator Associate, Google Associate Cloud Engineer.

Collaboration: Helpdesk staff provide the first response and accurate tickets; network administrators keep connectivity secure and available; Linux administrators operate internal services; and cloud administrators maintain SaaS and cloud resources. Shared alerts, change records, and escalation paths provide coordinated support.

Part 7 — Infrastructure Recommendations

1. Procure at least 300 Mbps symmetrical business fibre, a static IP, SLA, and LTE/5G failover. 2. Use ECC memory, redundant storage, TPM, remote management, and backed-up virtual machines for the server. 3. Apply a 3-2-1 strategy: production data, NAS snapshots, encrypted external drives off-site, and immutable cloud backup for Microsoft 365; test restores quarterly. 4. Use default-deny inter-VLAN rules, patches, centralized logs, least privilege, endpoint protection, and MFA for remote or privileged access. 5. Adopt unique long passphrases, password-manager use, breached-password blocklists, rate limiting, and compromise-driven password changes. 6. Reserve address, cabling, rack, and switch capacity for early growth.

Part 8 — Personal Reflection

This activity showed me that infrastructure planning is a business task as well as a technical one. The scenario made me consider users, sensitive data, downtime, and the documentation another technician needs. The most challenging part was separating departments without creating unnecessary complexity. VLANs and firewall rules provide a logical boundary for Finance and HR while retaining the services that each department needs. Planning before deployment identifies quantities, compatibility, addressing, cabling, backup locations, account controls, and recovery requirements while they are still inexpensive to change. This project helps me become a better system administrator because it practices a repeatable method: understand the organization, document assumptions, select appropriate technology, protect critical services, and explain decisions clearly to both management and technical colleagues. I also learned that an infrastructure plan is a living document. Once equipment is purchased, the inventory should be updated with real asset tags, serial numbers, warranty dates, assigned users, and configuration backups. When the company grows or changes its services, the network diagram and access rules should be reviewed instead of relying on memory. Maintaining those records will make troubleshooting faster and handovers between IT staff safer. This project therefore strengthened both my technical planning skills and my responsibility to maintain accurate documentation throughout a system’s lifecycle.

References

Microsoft Learn — Windows security features and edition requirements; Ubuntu Server documentation; Cisco — firewall guidance for small business; CISA — Require multifactor authentication; NIST SP 800-63B; AWS Certified Cloud Practitioner. Full hyperlinks are listed in README.md.